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(45) **Date of Patent:** Sep. 2, 2025

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Primary Examiner — Tan-Uyen T Ho

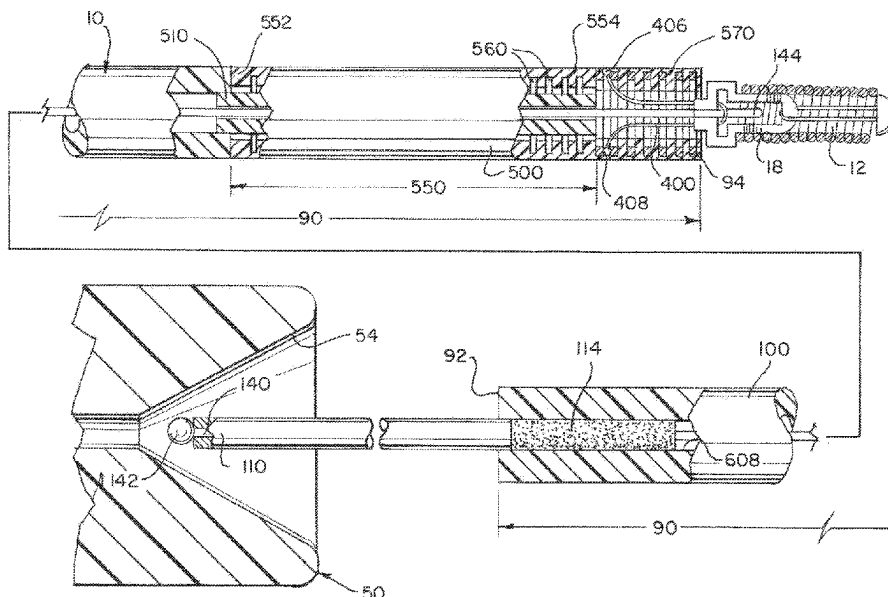
Assistant Examiner — Osama Nemer

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(57) **ABSTRACT**

A delivery system for deploying an implantable medical device to a target location of a body vessel can include a tubular body with a lumen extending therethrough, a pull wire extending through the lumen, and a proximal support tube disposed within the lumen and engaged to the pull wire such that proximal translation of the support tube causes proximal translation of the pull wire. The proximal support tube can include a marked distal section that is not visible to an operator of the delivery system prior to the release of the implant and that becomes visible when the proximal support tube is pulled proximally a sufficient distance to cause the delivery system to release the implant.

20 Claims, 7 Drawing Sheets



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FIG. 1

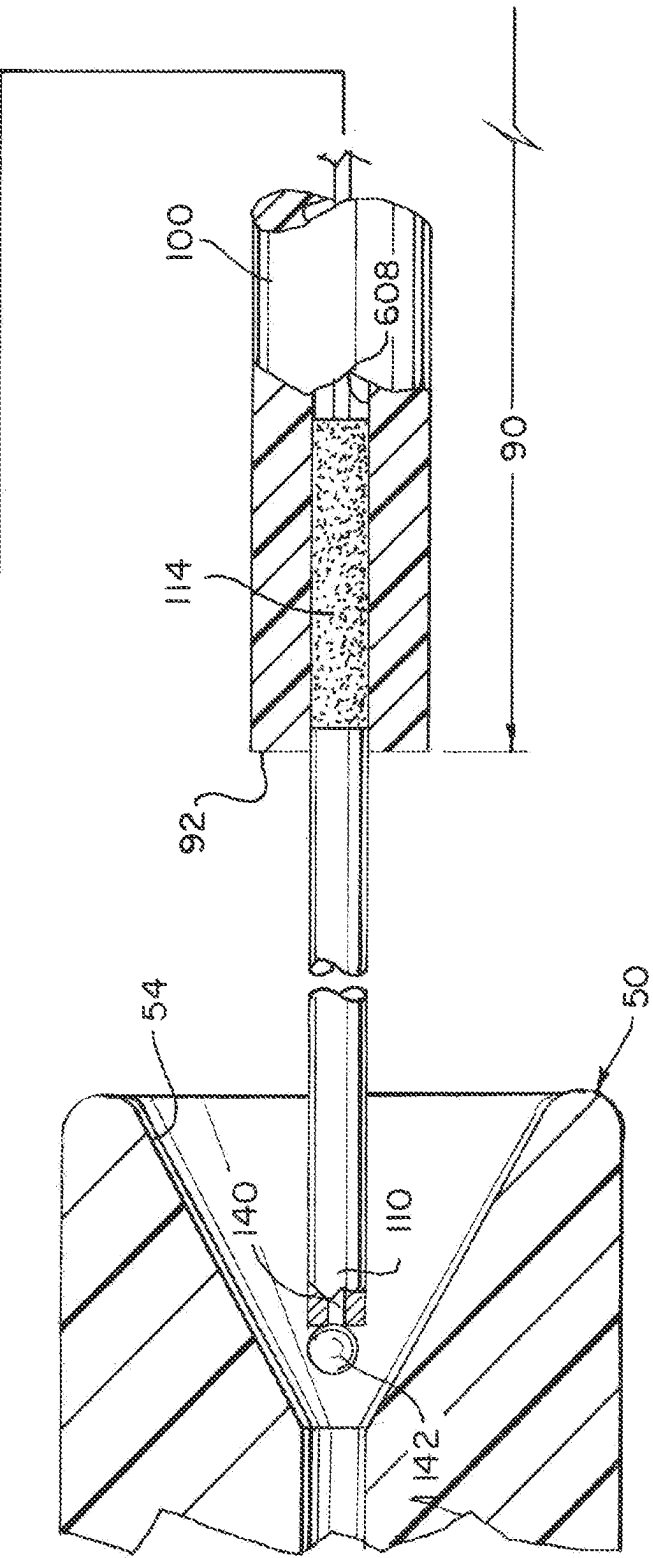
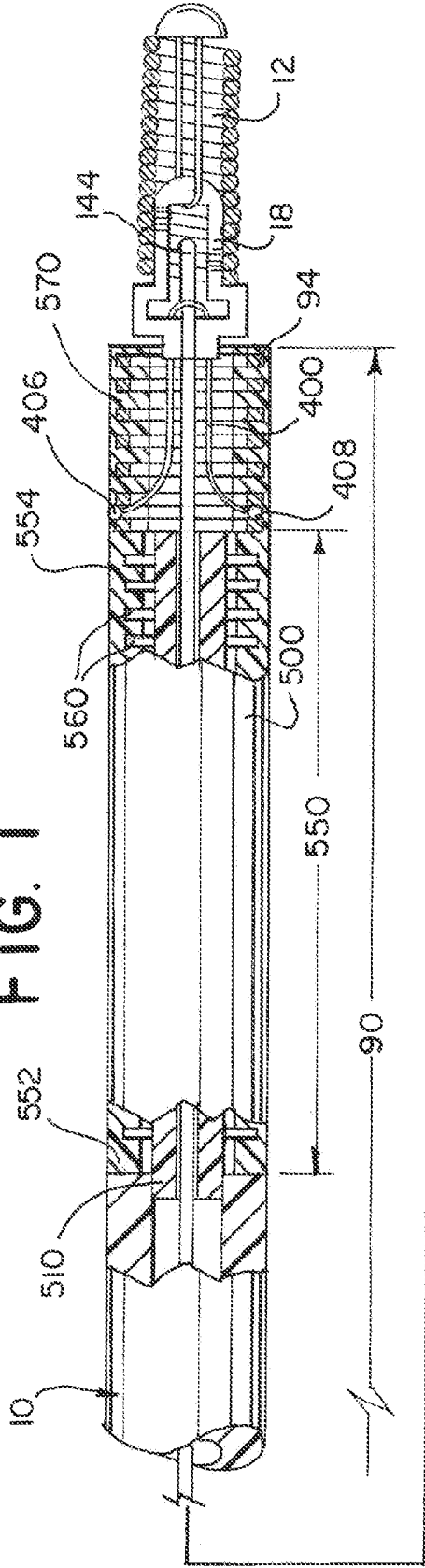


FIG. 2A

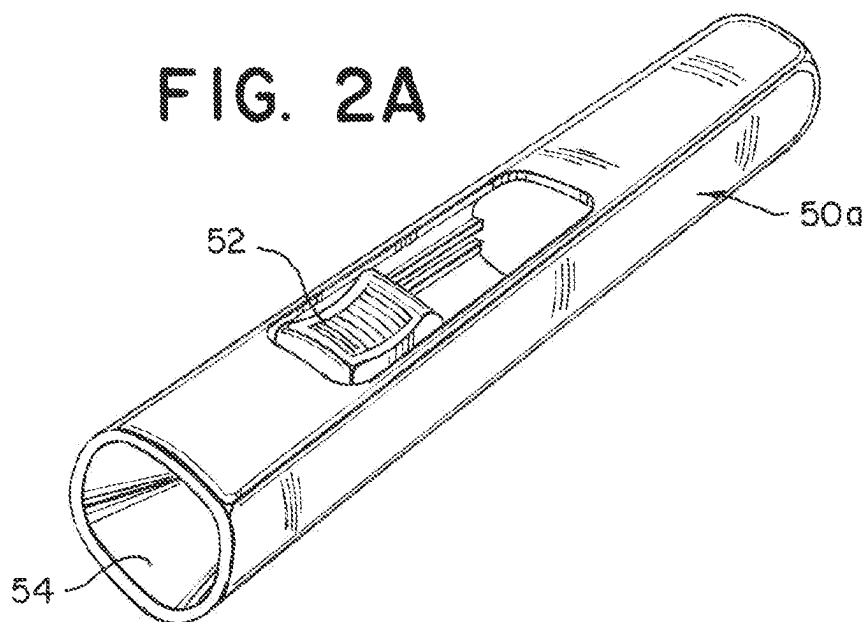


FIG. 2B

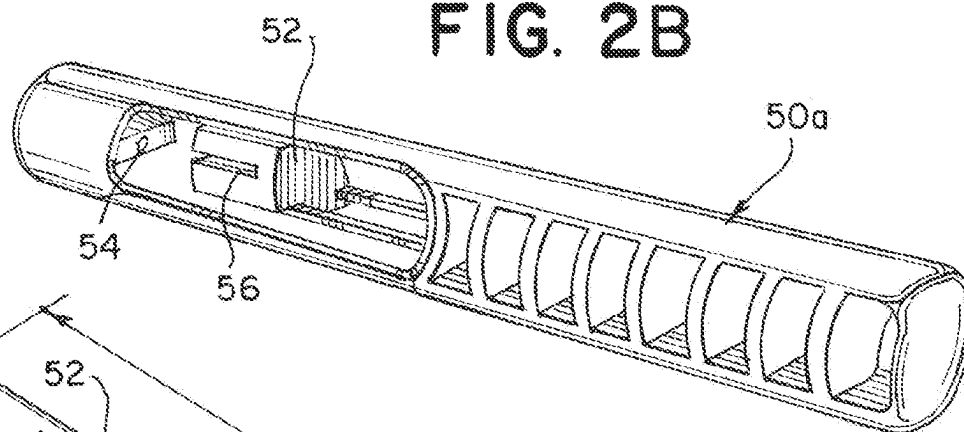


FIG. 2C

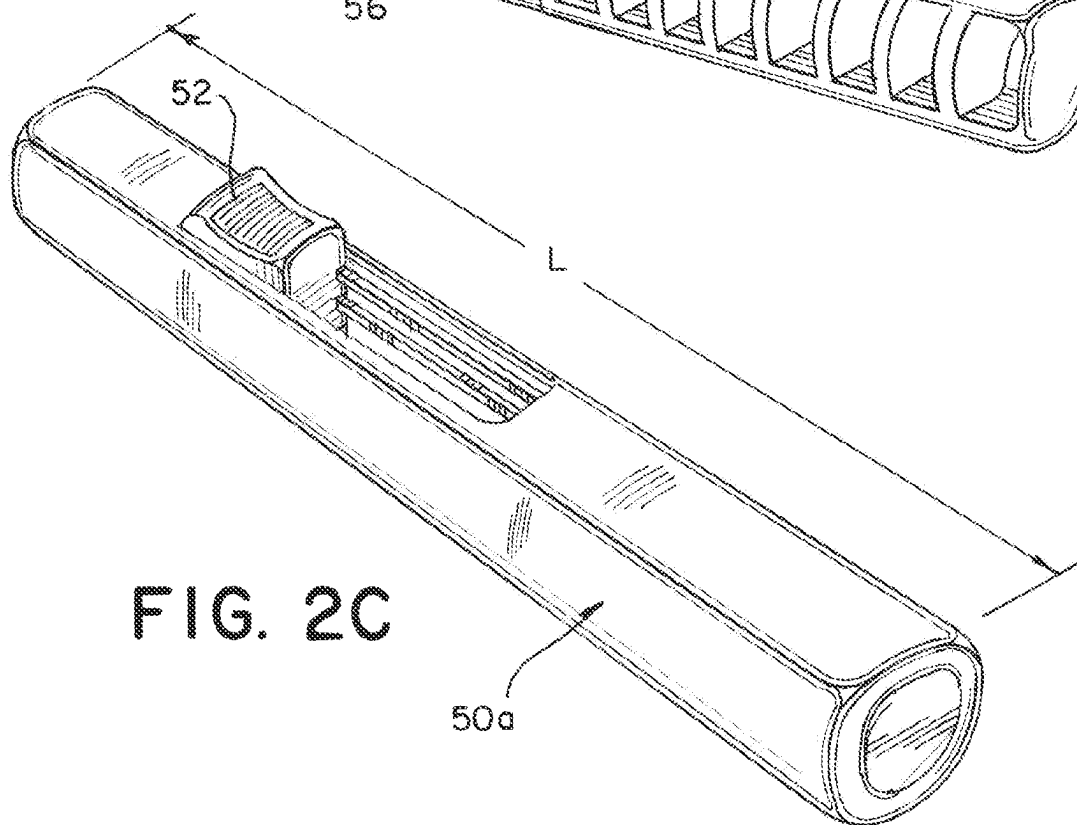


FIG. 3

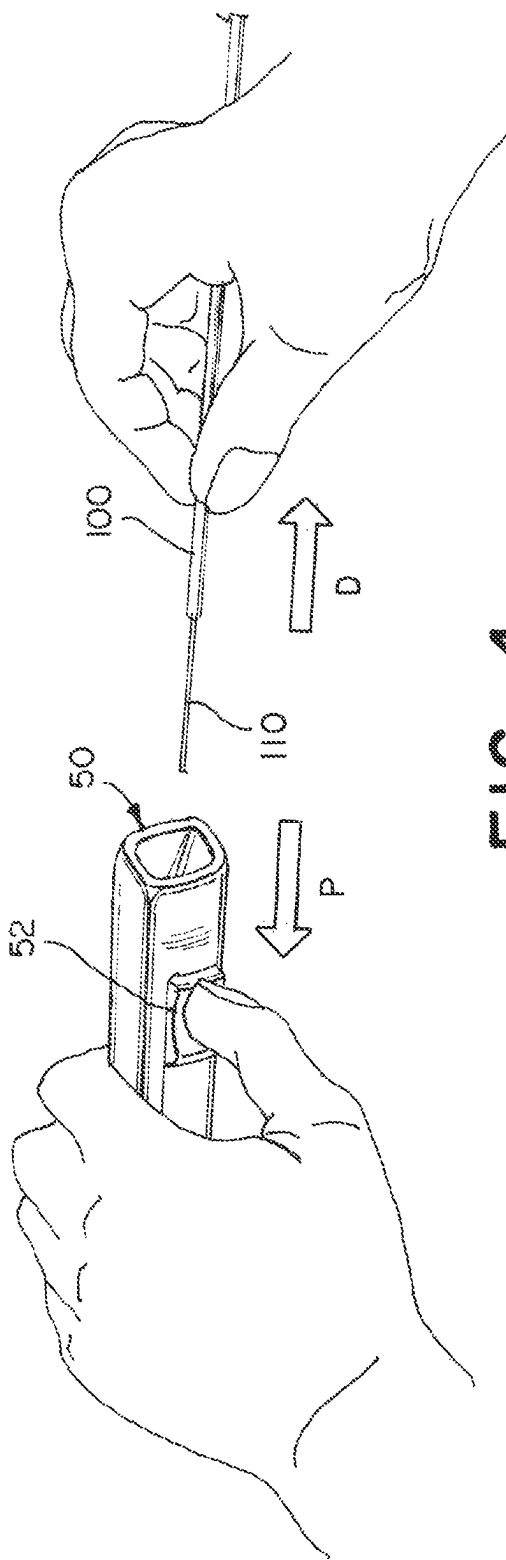


FIG. 4

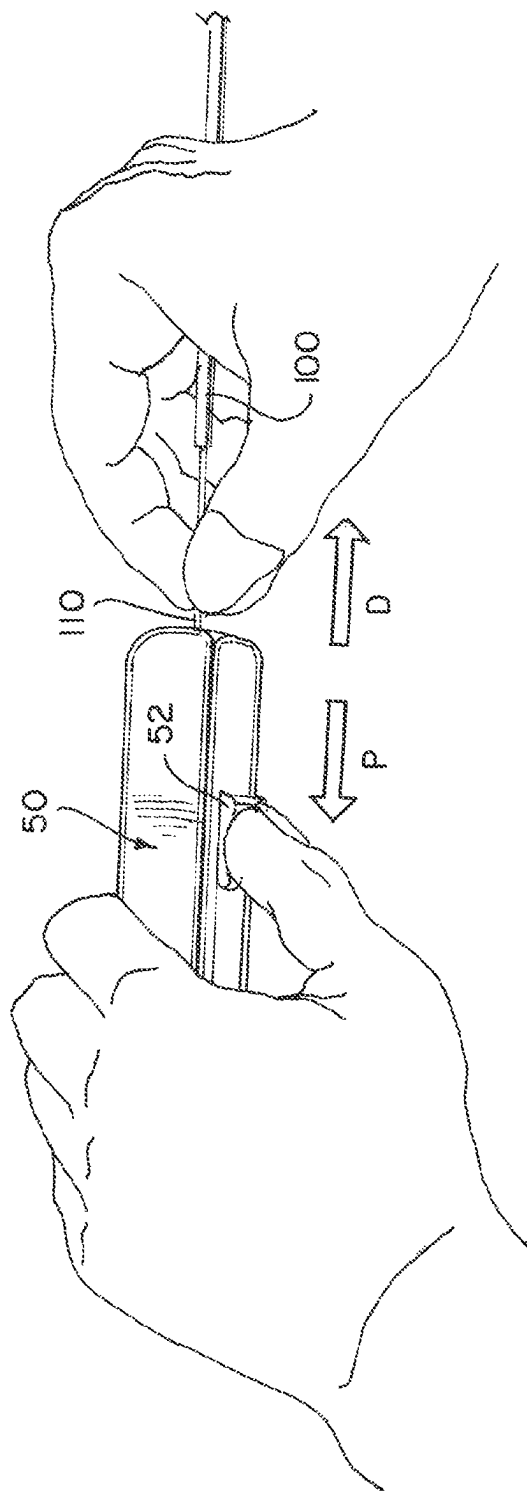


FIG. 5

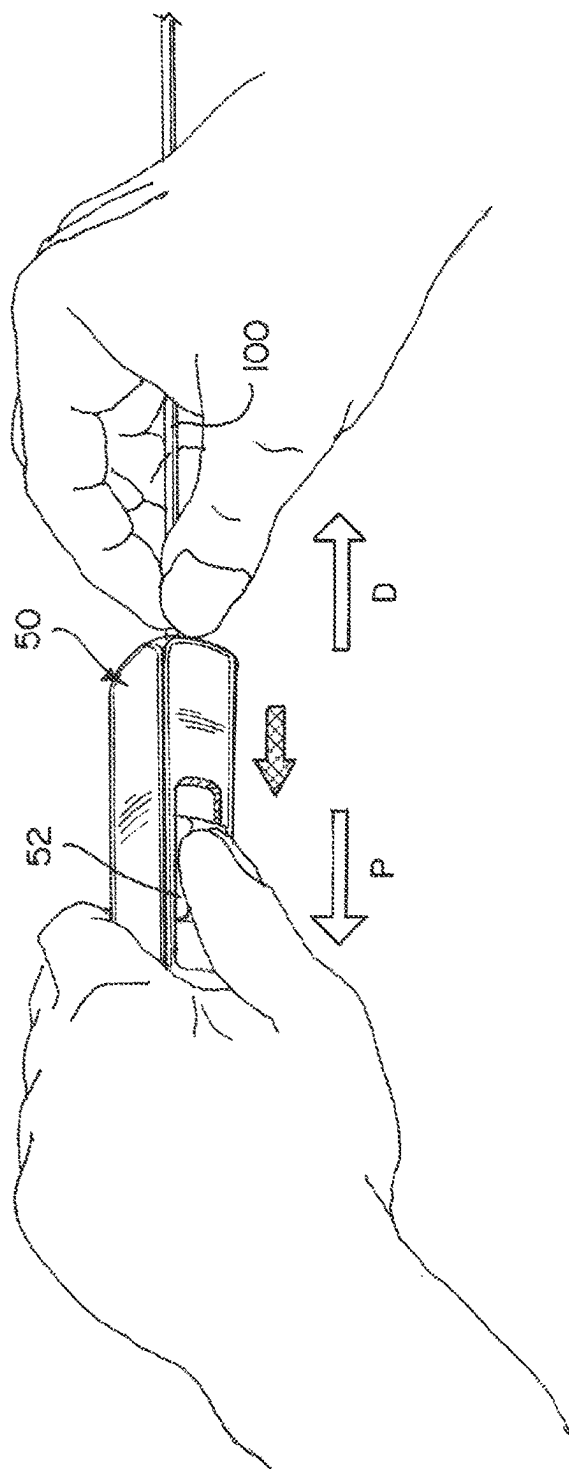
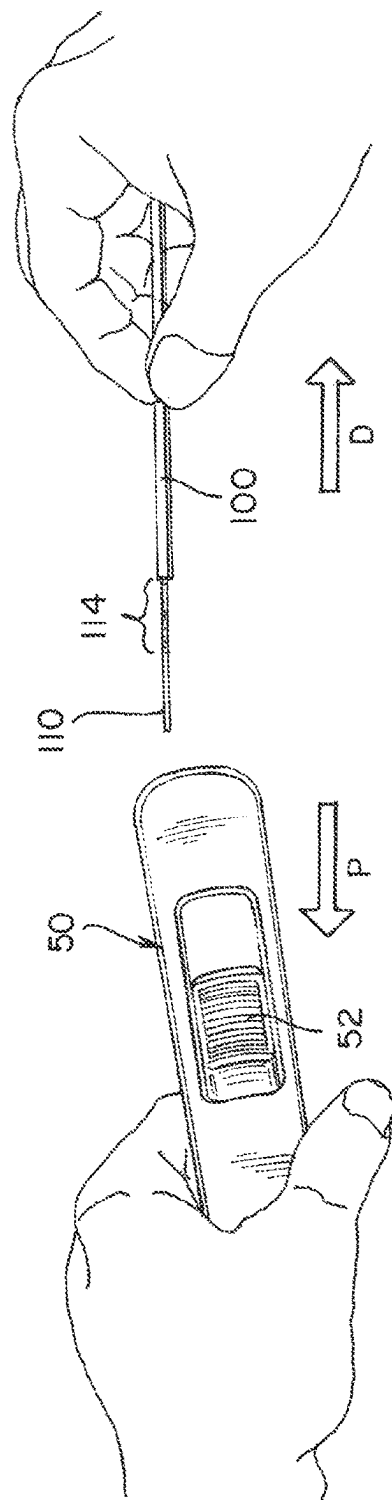
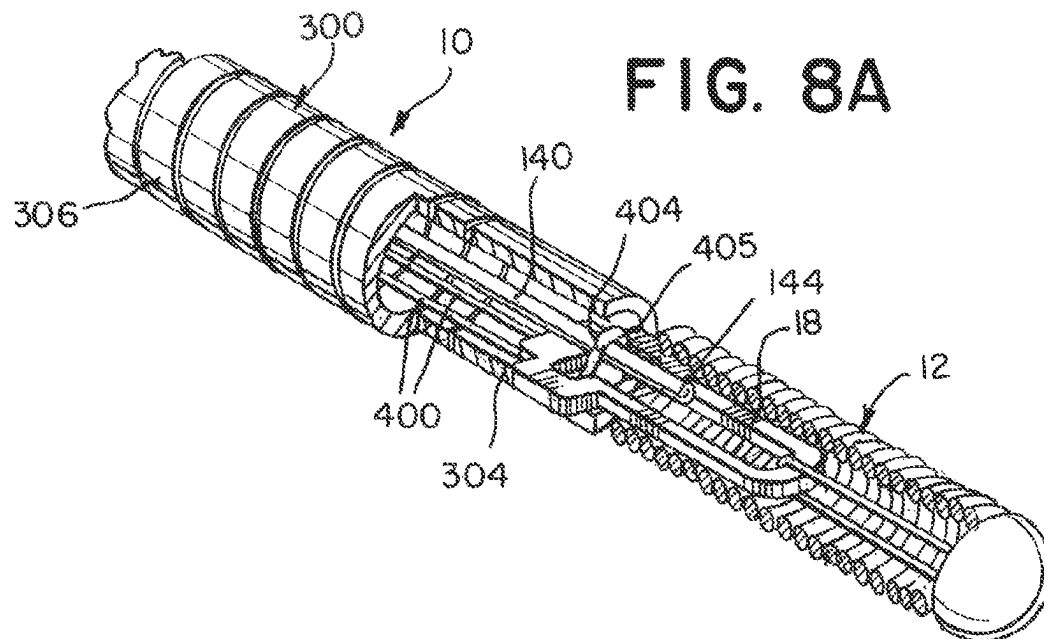
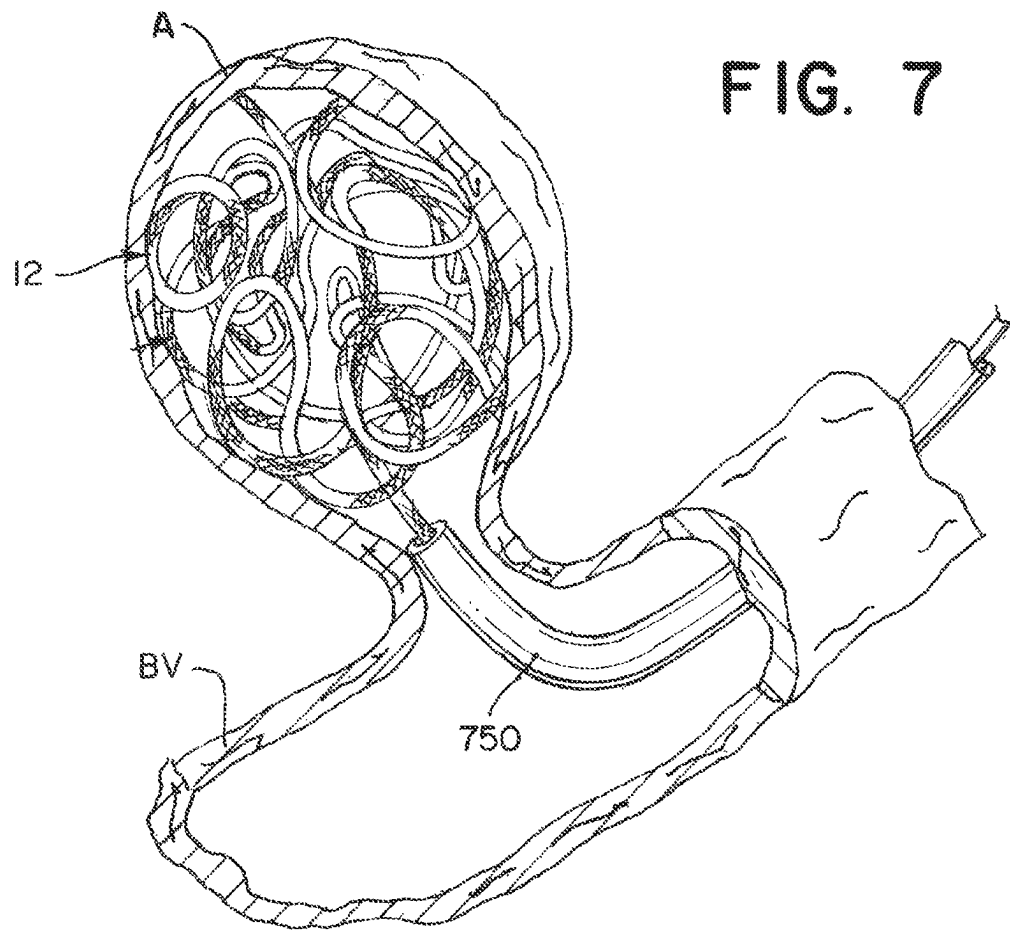


FIG. 6





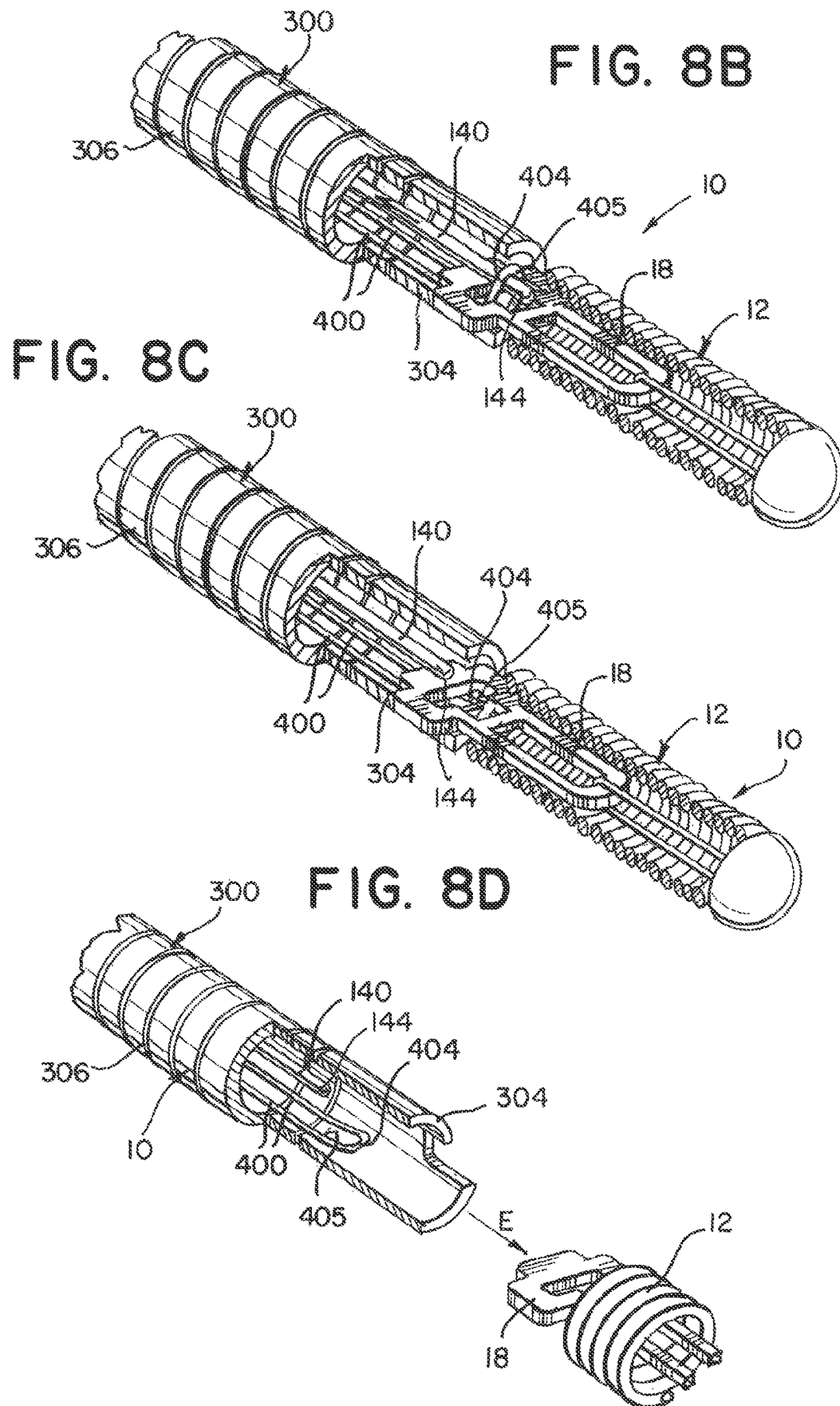
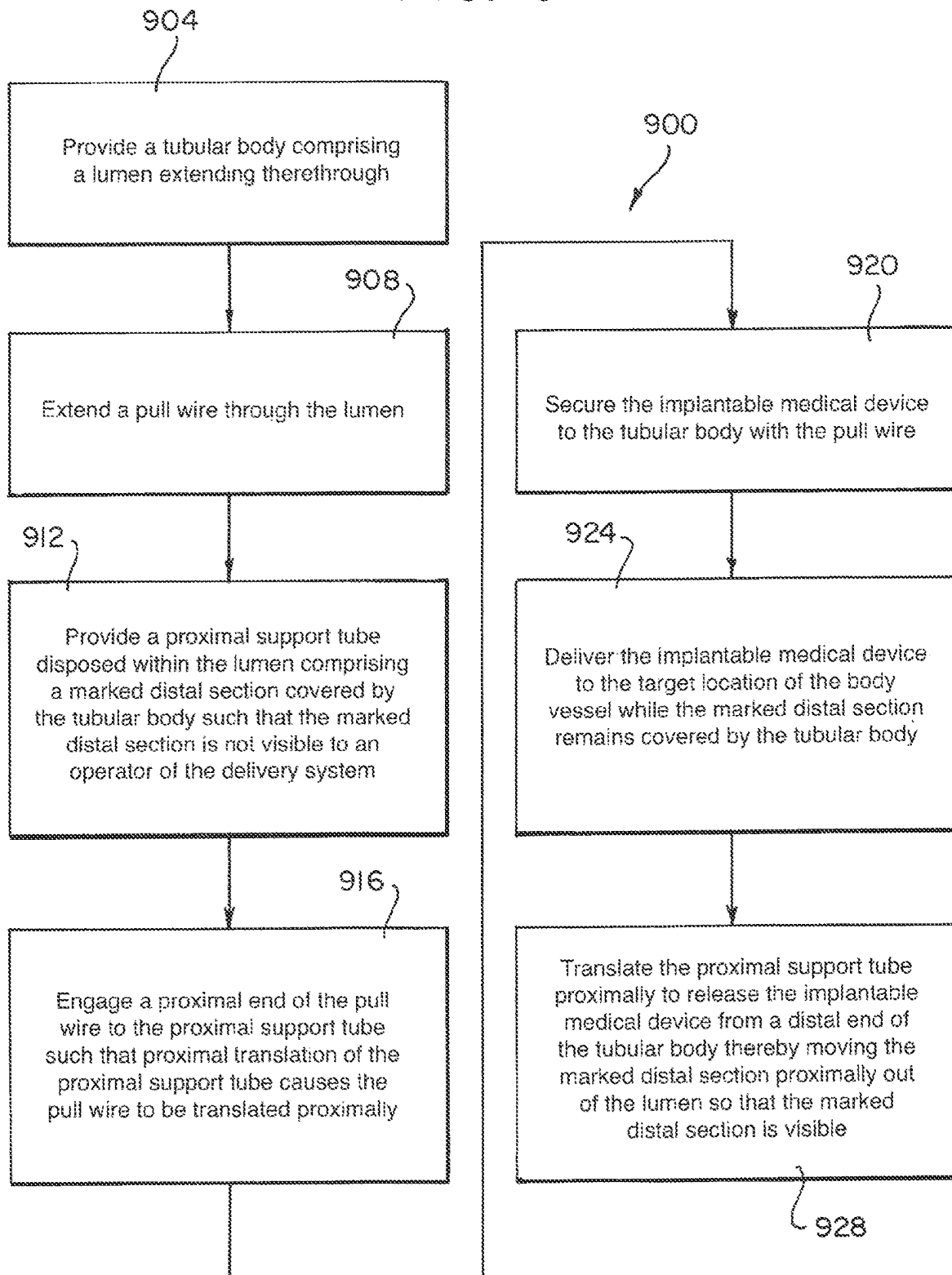


FIG. 9



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DETACHMENT INDICATOR FOR IMPLANT DEPLOYMENT

FIELD OF INVENTION

The present invention relate to aneurysm treatment devices and more particularly, to a delivery system for deploying an implantable medical device to a target location of a body vessel that includes a detachment indicator indicating when the implantable medical device has been deployed.

BACKGROUND

Numerous intravascular implant devices are known in the field. Many are deployed mechanically, via systems that combine one or more catheters and wires for delivery.

Examples of implants that can be delivered mechanically include embolic elements, stents, grafts, drug delivery implants, flow diverters, filters, stimulation leads, sensing leads, or other implantable structures delivered through a microcatheter. A physician desires feedback that a detachment cycle has been completed correctly. In most mechanical delivery systems, an inner tube is held within a lumen of a delivery system pusher and is attached to an internal wire that travels the length of the device. When pulled, the internal wire translates backwards to detach the implant. In traditional systems, an identifier that the detachment cycle has been completed is that the inner tube has been translated and thus increases in length. However, in traditional deployment systems, it is difficult for the physician to visually determine whether the implant has been deployed successfully because the small change in length is difficult to determine visually.

Accordingly, there is a need for a deployment system that has a detachment indicator indicating to the physician that the implant has been successfully deployed.

SUMMARY

Disclosed herein are various exemplary systems, devices, and methods of the present disclosure that can address the above needs. Examples can generally include a delivery system for deploying an implantable medical device to a target location of a body vessel. The delivery system can include a tubular body that includes a lumen extending therethrough. The delivery system can include a pull wire that extends through the lumen. The delivery system can include a proximal support tube that is disposed within the lumen. The proximal support tube can include a marked distal section that is covered by the tubular body such that the marked distal section is not visible to an operator of the delivery system. The proximal support tube can be configured to engage a proximal end of the pull wire such that proximal translation of the proximal support tube causes the pull wire to be translated proximally thereby releasing an implantable medical device from a distal end of the tubular body. The proximal support tube can be configured such that proximal translation of the proximal support tube resulting in release of the implantable medical device moves the marked distal section from the lumen of the tubular body so that the marked distal section is visible.

The delivery system can further include a loop wire that includes a loop opening at a distal portion of the loop wire positioned approximate the distal end of the tubular body. The loop wire and the pull wire are positioned to secure the implantable medical device to the delivery system.

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The implantable medical device can be deployed in response to proximal translation of the proximal support tube, which can cause a distal end of the pull wire to exit the loop opening, which can cause the loop wire and the pull wire to disengage the medical device from the delivery system.

The marked distal section can include a laser etching visible to the operator of the delivery system upon deployment of the implantable medical device.

The marked distal section can include an anodized portion of the proximal support tube visible to an operator of the delivery system upon deployment of the implantable medical device.

The marked distal section can include a printed or dipped portion of the proximal support tube visible to an operator of the delivery system upon deployment of the implantable medical device.

The marked distal section can be visibly distinguishable from a remainder of the proximal support tube.

A remainder of the proximal support tube can include a first color, and the marked distal section can include a second color distinguishable from the first color.

Proximal translation of the proximal support tube can be between approximately 6 millimeters and approximately 12 millimeters to expose the marked distal portion.

Proximal translation of the proximal support tube can be between approximately 6 millimeters and approximately 12 millimeters to deploy the implantable medical device.

The tubular body can include a proximal tube, a flexible tube distal to the proximal tube, and a distal tube distal to the flexible tube.

The flexible tube can include interference cuts.

The distal tube can further include a compressed distal portion positioned approximate the implantable medical device. The compressed portion can provide an elastic force against the implantable medical device to thereby facilitate the release of the implantable medical device from the delivery system.

The compressed distal portion can include a spiral cut portion of the distal tube.

In another aspect, a method for using a delivery system to deploy an implantable medical device to a target location of a body vessel is disclosed. The method can include providing a tubular body. The tubular body can include a lumen extending therethrough. The method can include extending a pull wire through the lumen. The method can include providing a proximal support tube that is disposed within the lumen. The proximal support tube can include a marked distal section that is covered by the tubular body such that the marked distal section is not visible to an operator of the delivery system. The method can include engaging a proximal end of the pull wire to the proximal support tube such that proximal translation of the proximal support tube causes the pull wire to be translated proximally. The method can include securing the implantable medical device to the tubular body with the pull wire. The method can include delivering the implantable medical device to the target location of the body vessel while the marked distal section remains covered by the tubular body. The method can include translating a proximal support tube proximally to release the implantable medical device from a distal end of the tubular body, thereby moving the marked distal section proximally out of the lumen so that the marked distal section is visible.

The marked distal section can include a laser etching visible to an operator of the delivery system upon deployment of the implantable medical device.

The marked distal section can include an anodized portion of the proximal support tube visible to an operator of the delivery system upon deployment of the implantable medical device.

The marked distal section can include a printed or dipped portion of the proximal support tube visible to an operator of the delivery system upon deployment of the implantable medical device.

A remainder of the proximal support tube can include a first color and the marked distal section can include a second color distinguishable from the first color.

Proximal translation of the proximal support tube of between approximately 6 millimeters and approximately 12 millimeters exposes the marked distal portion and is effective to deploy the implantable medical device.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and further aspects of this invention are further discussed with reference to the following description in conjunction with the accompanying drawings, in which like numerals indicate like structural elements and features in various figures. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating principles of the invention. The figures depict one or more implementations of the inventive devices, by way of example only, not by way of limitation.

FIG. 1 is an illustration of a delivery system and implant, according to aspects of the present invention.

FIG. 2A-2C illustrate a deployment handle and delivery system in an unconnected state.

FIG. 3 illustrates a deployment handle and delivery system in an unconnected state.

FIG. 4 illustrates a deployment handle and delivery system in a connected state.

FIG. 5 illustrates a deployment handle and delivery system as the deployment handle is used to deploy an implant from the delivery system.

FIG. 6 shows a deployment handle and delivery system in an unconnected state after the implant is deployed.

FIG. 7 is an illustration of embolic coils being positioned within an aneurysm according to aspects of the present invention.

FIGS. 8A-8D illustrate a sequence of steps for releasing an implant from the delivery system, according to aspects of the present invention.

FIG. 9 is a flowchart of an exemplary method of using the deployment handle and delivery system to deploy an implant.

DETAILED DESCRIPTION

The following description of certain examples of the invention should not be used to limit the scope of the present invention. The drawings, which are not necessarily to scale, depict selected embodiments and are not intended to limit the scope of the invention. The detailed description illustrates by way of example, not by way of limitation, the principles of the invention. Other examples, features, aspects, embodiments, and advantages of the invention will become apparent to those skilled in the pertinent art from the following description, which is by way of illustration, one of the best modes contemplated for carrying out the invention. As will be realized, the invention is capable of other different or equivalent aspects, all without departing from

the invention. Accordingly, the drawings and descriptions should be regarded as illustrative in nature and not restrictive.

Any one or more of the teachings, expressions, versions, examples, etc. described herein may be combined with any one or more of the other teachings, expressions, versions, examples, etc. that are described herein. The following-described teachings, expressions, versions, examples, etc. should therefore not be viewed in isolation relative to each other. Various suitable ways in which the teachings herein may be combined will be readily apparent to those skilled in the pertinent art in view of the teachings herein. Such modifications and variations are intended to be included within the scope of the claims.

As used herein, the terms “about” or “approximately” for any numerical values or ranges indicate a suitable dimensional tolerance that allows the part or collection of components to function for its intended purpose as described herein. More specifically, “about” or “approximately” may refer to the range of values $\pm 10\%$ of the recited value, e.g. “about 90%” may refer to the range of values from 81% to 99%. In addition, as used herein, the terms “patient,” “host,” “user,” and “subject” refer to any human or animal subject and are not intended to limit the systems or methods to human use, although use of the subject invention in a human patient represents a preferred embodiment.

Turning to the figures, FIG. 1 shows an example delivery system 10 for deploying an implantable medical device 12 to a target location of a body vessel. The example delivery system 10 can include a proximal tube 100, a flexible tube 550 positioned distal to the proximal tube 100, and a distal tube 570 positioned distal to the flexible tube 550. Flexible tube 550 has proximal end 552 and distal end 554. The proximal tube 100, flexible tube 550, and distal tube 570 may collectively have a lumen 608 extending therethrough. That is, lumen 608 extends from a proximal end of proximal tube 100, through flexible tube 550, and through distal tube 570. Collectively, proximal tube 100, flexible tube 550 and distal tube 570 can be referred to as a tubular body 90. The tubular body can include a proximal end 92 and a distal end 94, as shown in FIG. 1. Implant 12 can be positioned distal to the distal tube 570.

The flexible tube 550 can include a coiled wire, support coil that is attached at each end to the proximal tube 110 and the distal tube 570 respectively. Alternatively, the flexible tube 550 can include a laser cut hypotube including interference cuts 560 as illustrated in FIG. 1. In this example, the flexible tube 550 is preferably contiguous with the proximal tube 100 and the distal tube 570 such that the tubes 100, 550, 570 are formed from a single hypotube. The interference cuts 560 can provide a high level of flexibility to flexible tube 550. A sleeve 500 can surround the flexible tube 550 to provide a smooth outer surface of the delivery system 10 over the flexible tube 550.

An inner support tube 510 can be positioned within the lumen 608 of the flexible tube 550. The inner support tube 510 can be welded or otherwise attached to the inner sidewall of flexible tube 550. Inner support tube 510 can provide structural support to the flexible tube 550. In some examples, the inner support tube 510 can prevent the flexible tube 550 from elongating. The inner support tube 510 is optional if the interference cuts are made to allow the flexible tube 550 to have sufficient structural support in the absence of the support tube 510, or if other features are included to provide structural support to the flexible tube 550.

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A pull wire **140** can extend through the lumen **608** of delivery system **10**. Within lumen **608** of proximal tube **100** can be positioned a proximal support tube **110**. Proximal support tube **110** can be welded or otherwise attached to the inner sidewall of proximal tube **100**. Both proximal support tube **110** and inner support tube **510** can be considered as part of tubular body **90**. Proximal support tube **110** can be considered to be located on proximal end **92** of the tubular body **90**. The pull wire **140** can include a pull wire bead **142** located on a proximal end of pull wire **140**. The pull wire bead **142** can be sized such that the pull wire bead **142** is larger than the space between the proximal support tube **110** such that the pull wire bead **142** interferes with the proximal support tube **110** and proximal support tube **110** retains the pull wire bead **142** proximal to the proximal tube **100**. Alternatively, the proximal end of the pull wire can be welded to the proximal tube **110** or otherwise engaged to the proximal tube **110** as understood by a person skilled in the pertinent art.

A loop wire opening **405** of loop wire **400** (also illustrated with respect to FIGS. **8A-8D**) can be located at a distal portion **404** of loop wire **400** and can extend through a locking member **18** of implant **12**. The distal end **144** of pull wire **140** can be positioned through the loop wire opening **405** of loop wire **400** to secure implant **12** to the delivery system **10**. Loop wire proximal ends **406**, **408** can be attached to the distal tube **570** as illustrated. Additionally or alternatively, one or both of the loop wire proximal ends **406**, **408** can be affixed to the proximal tube **100** to provide structural support to the flexible tube **550**. Additionally, or alternatively, the loop wire proximal ends **406**, **408** can be attached to the pull wire **140**. In this example, the loop wire **400** can be made of an elastic material, such that proximal translation of the pull wire **140** can cause the loop wire **400** to stretch and become tensioned, which provides a resistant force against further proximal translation of pull wire **140**. The loop wire **400** can be effective to prevent premature detachment of implant **12** from delivery system **10**.

The proximal support tube **110** can include a marked distal section **114** that is located within the proximal tube **100**. The marked distal section **114** can be configured to be hidden from view of an operator of the delivery system **10** while implant **12** is delivered to a target location within a patient. When the pull wire **140** is pulled proximally a distance sufficient to detach implant **12**, the proximal support tube **110** can be configured to translate proximally, thereby causing the marked distal section **114** to exit the proximal tube **100** and become visible to an operator of the delivery system **10**.

When the delivery system **10** is assembled, the flexible tube **550** and sleeve **500** can be more flexible than the distal tube **570** and the proximal tube **100**. One way to measure flexibility is to perform a three-point bend test wherein a portion of the delivery system **10** is held fixed at two end points, a force is applied perpendicularly to the delivery system **10** centrally between the points, and flexibility is quantified by the length of deflection of the delivery system **10** caused by the force. When measured in this way, in some examples, the flexible tube **550** and sleeve **500** can be about 1.5 times more flexible than the distal tube **570** and about 20 times more flexible than the proximal tube **100**. That is, when the three-point test is performed identically on the three sections **100**, **550**, and **570**, the flexible tube **550** can deflect over a length that is about 1.5 times the deflection of the distal tube **570** and about 20 times the length of deflection of the proximal tube **100**. Flexibility can be measured

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in other ways as would be appreciated and understood by a person having pertinent skill in the art.

According to some embodiments, the distal tube **570** can include a compressible distal portion **300** (as shown in greater detail in FIGS. **8A** through **8D**), which can be compressed and formed from a spiral-cut portion of the distal tube **570**, formed by a laser cutting operation. Additionally, or alternatively, the compressible distal portion **300** can be formed of a wound wire, spiral ribbon, or other arrangement allowing axial adjustment according to the present invention. Preferably, compressible distal portion **300** is in the elongated condition at rest and automatically or resiliently returns to the elongated condition from a compressed condition, unless otherwise constrained. In some embodiments, the loop wire **400** is effective to hold the compressed distal portion **300** in compression while the implant **12** is delivered to the treatment site. Alternatively, the distal tube **570** can lack the compressible spring structure **300**.

Pull wire **140** can be constructed out of any suitable material, for example, pull wire **140** can be constructed of stainless steel or memory shape material, such as nitinol. According to some embodiments, pull wire **140** can additionally be coated with polytetrafluoroethylene (PTFE).

FIGS. **2A** through **2C** are illustration of an exemplary handle member **50**, according to aspects of the present disclosure. A deployment handle **50** can include components as described in pending U.S. application Ser. No. 17/564,764, published as U.S. Patent Application Publication No. US 2023/0200819 A1 on Jun. 29, 2023, which is herein incorporated by reference as if included in its entirety. Numerous types of handle members can be used to retract the support tube **110** and pull wire **140** as understood by a person skilled in the pertinent art.

FIG. **2A** is a top view perspective illustration of handle member **50** from the distal end, FIG. **2B** is a bottom cut-away view of handle member **50** from the proximal end, and FIG. **2C** is a top view perspective illustration of handle member **50** from the proximal end. The exemplary handle member **50** can include a handle button **52**, an aperture **54**, and a handle channel **56**. The handle member may have a length **L** extending from a proximal end of the handle member **50** to the distal end of the handle member **50**. According to some aspects of the present disclosure, the length **L** of handle member **50** can be approximately between 10 cm and 20 cm.

The handle channel **56**, found on the bottom of handle member **50a**, can be configured to engage with proximal support tube **110** and can have a length sufficient to engage to the length of proximal support tube **110**. Aperture **54** has a diameter sufficient to fit proximal support tube **110** through. Proximal support tube **110** is configured to pass through aperture **54** when engaging to handle member **50** at handle channel **56**.

Handle button **52** can be configured to be pressed by an operator of the delivery system **10**. When handle button **52** is pressed (e.g., towards a bottom surface of handle member **50**), the proximal support tube **110** can be pressed against the top surface of handle channel **56**, which can cause the support tube **110** and pull wire **140** to be retracted, thereby deploying the implant **12**. When the proximal end of the delivery system **10** is removed from the handle **50**, the marked distal section **114** of the support tube **110** can be visible to the operator. The marked distal section **114** provides a visible indication to the operator that the delivery system **10** has been properly manipulated to deploy the implant **12**.

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FIGS. 3-6 illustrate use of the delivery system 10 with another example handle 50. The operator can slide handle button 52 proximally to thereby detach the implant from the deployment system 10 and expose the marked distal section 114 of the proximal support tube 110 to provide a visual indication to the operator that the implant 12 has been successfully detached from the deployment system 10 and deployed in the target location of a patient's vasculature.

FIG. 3 illustrates a deployment handle 50 and delivery system 10 in an unconnected state with the handle 50. The proximal direction (P) and distal direction (D) are indicated in the illustration. The proximal tube 100 can have a proximal support tube 110 extending in a proximal direction (P) therefrom. The proximal support tube 110 can be configured to be inserted into aperture 54 of handle member 50. As shown, handle member 50 can include a handle button 52, which allows the operator of the delivery system 10 and the handle member 50 to engage the proximal support tube 110 against the handle channel 56 such that the proximal support tube 110 can be translated proximally by the operator slide handle button 52 in a proximal direction (P) while handle button 52 is depressed. Proximal translation of the proximal support tube 110 can cause the pull wire 140 to be translated proximally to thereby release the implant 12 from the delivery system 10. As shown in FIG. 3, the marked distal section 114 of the proximal support tube 110 is not visible, because the implant 12 has not yet been deployed from the delivery system 10.

FIG. 4 illustrates a deployment handle and delivery system in a connected state. As shown, an operator of the handle 50 and delivery system 10 can connect the proximal support tube 110 that extends from the proximal tube 100 by inserting the proximal support tube 110 into the aperture 54 of handle member 50. Proximal support tube 110 can be inserted into aperture 54 of handle member 50 until the proximal support tube 110 extends fully through a handle channel similar to handle channel 56, as described with respect to FIGS. 2A through 2C.

FIG. 5 illustrates a deployment handle and delivery system as the deployment handle is used to deploy an implant 12 from the delivery system 10. After the proximal support tube 110 is inserted through aperture 54 and fully extended into handle channel 56, the operator can slide the handle button 52 proximally. As handle button 52 is slid proximally, the proximal support tube 110 can slide proximally. As proximal support tube 110 slides proximally, the pull wire 140 can be proximally translated with the proximal translation of the proximal support tube 110. The proximal translation of pull wire 140 can deploy the implant 12 from delivery system 10. In some examples, a proximal shift of between approximately 6 mm and approximately 12 mm can be effective to deploy the implant 12 from the delivery system 10. In some examples, movement of handle button 52 proximally by between approximately 6 mm and approximately 12 mm may cause the proximal support tube 110 and the pull wire 140 to translate proximally by between approximately 6 mm and approximately 12 mm, which can cause the implant 12 to be deployed from delivery system 10.

Alternatively, the handle 50 may include a shuttle system to move the proximal support tube 110 a greater distance than the distance through which the handle button 52 travels.

FIG. 6 shows a deployment handle and delivery system in an unconnected state after the implant 12 is deployed. After the proximal support tube 110 and pull wire 140 are translated proximally to deploy implant 12 from delivery system 10, the marked distal section 114 can be exposed to an

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operator of the handle 50 and delivery system 10. In some examples, the marked distal section 114 can be created by laser etching a distal portion of the proximal support tube 110 in order to create the marked distal section 114 that is visually distinct from the remainder of the proximal support tube 110. In some examples, the marked distal section 114 can be created by an anodizing process that is applied to a distal portion of the proximal support tube 110 in order to create the marked distal section 114 that is visually distinct from the remainder of the proximal support tube 110. In some examples, the marked distal section 114 can be created by applying a color printing process to a distal portion of the proximal support tube 110 in order to create the marked distal section 114 that is visually distinct from the remainder of the proximal support tube 110. In some examples, the marked distal section 114 can be created by dipping a distal portion of the proximal support tube 110 in a respective paint color in order to create the marked distal section 114 that is visually distinct from the remainder of the proximal support tube 110. In some examples, the marked distal section 114 can be a first color, while the remainder of the proximal support tube can be a second color that is visually distinct from the first color.

FIG. 7 is an illustration of embolic implant 12 being delivered through catheter 750 and positioned within an aneurysm A on a blood vessel BV. The implant 12 can loop and bend with the aneurysm sac to form a thrombotic mass. The implant can loop back on themselves and/or loop next to other implants. As the aneurysm A becomes increasingly packed, overlapping portions of the implant 12 can press into each other. As illustrated, the implant 12 includes an embolic coil. Numerous types of implants can be deployed by a pull wire mechanism as understood by a person skilled in the pertinent art. For instance, implants can include an embolic braid, a stent, or flow diverter structure.

FIGS. 8A-8D illustrate a time sequence of steps for releasing an embolic implant 12 from a delivery system 10. The delivery system 10 and implant 12 can be configured such as illustrated in the previous figures and as otherwise described herein. FIG. 8A illustrates an engagement system including the loop wire 400 and pull wire 140 locked into a locking portion 18 of the medical device 12, thereby forming a sub-assembly. The spiral cuts 306 of the compressible distal portion 300 can be compressed and the loop wire 400 opening 405 at a distal end 404 of the loop wire 400 can be placed through the locking portion 18. When the pull wire 140 is put through the opening 405 the medical device 12 is now secure. FIG. 8B illustrates the pull wire 140 being drawn proximally to begin the release sequence for the medical device 12. FIG. 8C illustrates the instant the distal end 144 of the pull wire 140 exits the opening 405 and the pull wire 140 is pulled free of the loop wire 400. The distal end 404 of the loop wire 400 falls away and disengaged from the locking portion 18. As can be seen, there is now nothing holding the implant 12 to the detachment system 10. FIG. 8D illustrates the end of the release sequence. Here, the compressible distal portion 300 has extended/returned to its original shape and "sprung" forward. An elastic force E is imparted by the distal end 304 of the compressible distal portion 300 to the medical device 12 to "push" it away to ensure a clean separation and delivery of the medical device 12.

The compressible distal portion 306 can have a difference in length (distance of compression) when measured in the compressed configuration and the original, uncompressed configuration of about 0.5 mm to about 0.75 mm. Greater elastic force E can be achieved by using a greater distance

of compression. The distance of compression can be determined by the sizing of the loop wire **400**, the shape of the locking portion **18**, and the shape of the distal end **304** of the compressible distal portion **300**.

FIG. 9 is a flowchart of an exemplary method **900** of using the deployment handle and delivery system to deploy an implant. In block **904**, the method can include providing a tubular body **90** that includes a lumen **608** extending there-through. In block **908**, the method can include extending pull wire **140** through the lumen **608**. In block **912**, the method can include providing a proximal support tube **110** that disposed within lumen **608**. The proximal support tube **110** can include a marked distal section **114** that is covered by the tubular body **90**. The marked distal section **114** can be invisible to an operator of the delivery system **10**.

In block **916**, the method can include engaging a proximal end **142** of the pull wire **140** to the proximal support tube **110**. Proximal translation of the proximal support tube **110** can cause the pull **140** to be translated proximally. In block **920**, the method can include securing the implantable medical device **12** to the tubular body **90** with the pull wire **140**.

In block **924**, the method can include delivering the implantable medical device **12** to the target location of the body vessel. While delivering the implantable medical device, the marked distal section **114** can remain covered by the tubular body **90** such that it is not visible to the operator of the delivery system.

In block **928**, the method can include translating the proximal support tube **110** proximally to release the implantable medical device **12** from a distal end **94** of the tubular body **90**. The marked distal section **114** can be moved proximally out of the lumen **608** such that the marked distal section **114** becomes visible to an operator of the delivery system. The marked distal section can thereby effectively indicate that the implantable medical device **12** has been successfully deployed at the target location of the body vessel.

The descriptions contained herein are examples of embodiments of the invention and are not intended in any way to limit the scope of the invention. As described herein, the invention contemplates many variations and modifications of the implantation system and associated methods, including alternative geometries of system components, alternative materials, additional or alternative method steps, etc. Modifications apparent to those skilled in the pertinent art are intended to be within the scope of the claims which follow.

The invention claimed is:

1. A delivery system for deploying an implantable medical device to a target location of a body vessel, the delivery system comprising:

- a handle comprising an aperture, a handle channel, and a handle button;
- a tubular body comprising a lumen extending there-through;
- a pull wire extending through the lumen; and
- a proximal support tube disposed within the lumen, a distal portion of the proximal support tube comprising a marked distal section covered by the tubular body such that the marked distal section is not visible to an operator of the delivery system, wherein the proximal support tube engages a proximal end of the pull wire such that proximal translation of the proximal support tube causes the pull wire to be translated proximally thereby releasing the implantable medical device from a distal end of the tubular body,

wherein the proximal support tube is inserted into the aperture of the handle, and interacts with the handle button such that proximal movement of the handle button results in proximal translation of the proximal support tube,

wherein at least a proximal portion of a length of the proximal support tube extends through the handle channel,

wherein proximal translation of the proximal support tube and the handle button results in release of the implantable medical device and moves the marked distal section proximally from the lumen of the tubular body into the handle channel so that the marked distal section is visible only when the handle is completely removed from the tubular body and the proximal support tube, and

wherein the handle is completely removed from the tubular body and the proximal support tube after the release of the implantable medical device.

2. The delivery system of claim 1, further comprising: a loop wire comprising a loop opening at a distal portion of the loop wire positioned approximate the distal end of the tubular body, wherein the loop wire and the pull wire are positioned to secure the implantable medical device to the delivery system.

3. The delivery system of claim 2, wherein the implantable medical device is deployed in response to proximal translation of the proximal support tube, which causes a distal end of the pull wire to exit the loop opening, which causes the loop wire and the pull wire to disengage the implantable medical device from the delivery system.

4. The delivery system of claim 1, wherein the marked distal section comprises a laser etching visible to the operator of the delivery system upon deployment of the implantable medical device upon complete removal of the handle from the tubular body and proximal support tube.

5. The delivery system of claim 1, wherein the marked distal section comprises an anodized portion of the proximal support tube visible to the operator of the delivery system upon deployment of the implantable medical device upon complete removal of the handle from the tubular body and proximal support tube.

6. The delivery system of claim 1, wherein the marked distal section comprises a printed or dipped portion of the proximal support tube visible to the operator of the delivery system upon deployment of the implantable medical device upon complete removal of the handle from the tubular body and proximal support tube.

7. The delivery system of claim 1, wherein the marked distal section is visibly distinguishable from a remainder of the proximal support tube.

8. The delivery system of claim 1, wherein a remainder of the proximal support tube comprises a first color, and the marked distal section comprises a second color distinguishable from the first color.

9. The delivery system of claim 1, wherein proximal translation of the proximal support tube of between approximately 6 millimeters and approximately 12 millimeters exposes the marked distal section upon complete removal of the handle from the tubular body and proximal support tube.

10. The delivery system of claim 1, wherein proximal translation of the proximal support tube between approximately 6 millimeters and approximately 12 millimeters is effective to deploy the implantable medical device.

11. The delivery system of claim 1, the tubular body further comprising:

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a proximal tube;
a flexible tube distal to the proximal tube; and
a distal tube distal to the flexible tube.

12. The delivery system of claim 11, wherein the flexible tube comprises interference cuts.

13. The delivery system of claim 11, wherein:
the distal tube further comprises a compressed distal portion positioned approximate the implantable medical device; and

the compressed distal portion provides an elastic force against the implantable medical device thereby facilitating the release of the implantable medical device from the delivery system.

14. The delivery system of claim 13, wherein the compressed distal portion further comprises a spiral cut portion of the distal tube.

15. A method for using a delivery system to deploy an implantable medical device to a target location of a body vessel, the method comprising:

providing a tubular body comprising a lumen extending therethrough;

extending a pull wire through the lumen;

providing a proximal support tube disposed within the lumen, the proximal support tube comprising a marked distal section covered by the tubular body such that the marked distal section is not visible to an operator of the delivery system;

engaging a proximal end of the pull wire to the proximal support tube such that proximal translation of the proximal support tube causes the pull wire to be translated proximally;

engaging a proximal end of the proximal support tube to a handle via an aperture in the handle;

engaging at least a proximal portion of a length of the proximal support tube with a handle channel in the handle;

securing the implantable medical device to the tubular body with the pull wire;

delivering the implantable medical device to the target location of the body vessel while the marked distal section remains covered by the tubular body;

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translating the proximal support tube proximally by proximally moving a handle button on the handle to release the implantable medical device from a distal end of the tubular body, thereby moving the marked distal section proximally out of the lumen into the handle channel so that the marked distal section is visible only when the handle is completely removed from the tubular body and proximal support tube; and

removing the handle completely from the tubular body and proximal support tube after the release of the implantable medical device, thereby exposing the marked distal section.

16. The method of claim 15, wherein the marked distal section comprises a laser etching visible to the operator of the delivery system upon deployment of the implantable medical device and complete removal of the handle from the tubular body and proximal support tube.

17. The method of claim 15, wherein the marked distal section comprises an anodized portion of the proximal support tube visible to the operator of the delivery system upon deployment of the implantable medical device and complete removal of the handle from the tubular body and proximal support tube.

18. The method of claim 15, wherein the marked distal section comprises a printed or dipped portion of the proximal support tube visible to the operator of the delivery system upon deployment of the implantable medical device and complete removal of the handle from the tubular body and proximal support tube.

19. The method of claim 15, wherein a remainder of the proximal support tube comprises a first color, and the marked distal section comprises a second color distinguishable from the first color.

20. The method of claim 15, wherein proximal translation of the proximal support tube of between approximately 6 millimeters and approximately 12 millimeters exposes the marked distal section upon complete removal of the handle from the tubular body and proximal support tube and is effective to deploy the implantable medical device.

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